

Element J

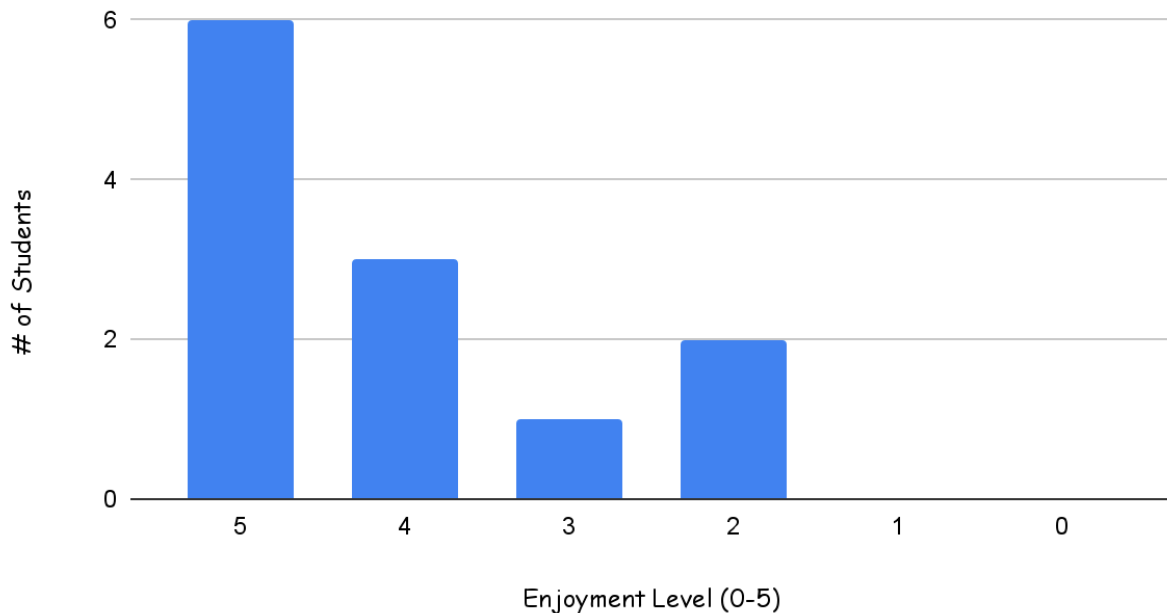
Documentation:

(involvement of stakeholders)

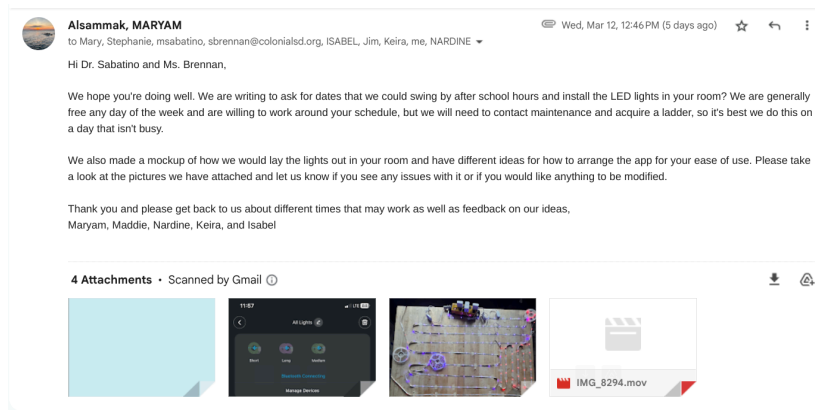
Stakeholders (Kids in the class):

We visited the class three times to get to know the kids and how the class is run. We went around doing our best to interact with the students and finding their likes and dislikes. We also looked in the sensory room to see how that was run and how the kids used that because our LED lights are going to give a sensory experience.

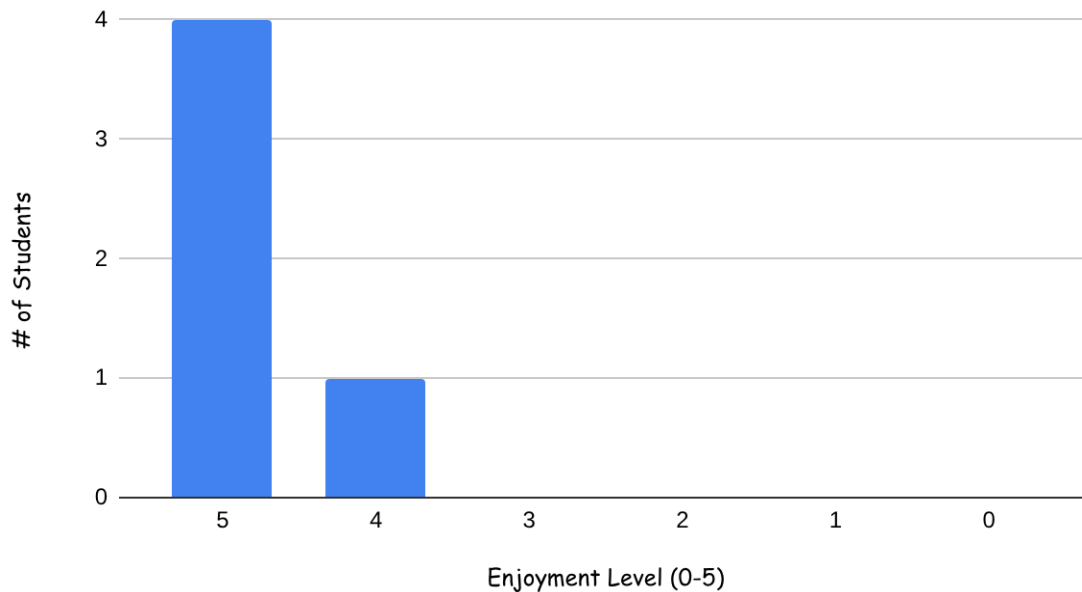
Student's Feedback



Stakeholders (Ms. Brennan and Dr. Sabatino):



Teacher's Feedback



Our team regularly checked in with Ms. Brennan and Dr. Sabatino in order to update and ensure that their needs were always being met and kept into consideration. Oftentimes, we reached out to Ms. Brennan to schedule hours we could come in to install the lights. From time to time, we would encounter difficulties during scheduling due to our busy strict high school -student times and her availability as a teacher at Whitemarsh.

Mechanical Engineer (Mr. Muscarella):

After reviewing our final product, we spoke with a mechanical engineer who suggested that we could have done some investigation into the intensity and brightness of the lights. He also said we could have done testing to determine what placement was best for the lights in relation to the windows. He also said we could have done more research about issues with seizures and safety for kids. He recommended we could have a warning label for teachers to make sure they don't set the lights to blink above a certain speed.

Biomedical Engineer (Dr. Sarver):

After reviewing our final product, Dr. Sarver felt that we hadn't yet achieved all of our goals. In general, we did come out of the project with what we said we would have—an LED light system. But we didn't do much to differentiate it from any other LED light system. The positive part of the lights was that we were able to separate them into zones and provide them with an app that teachers could use to customize to their own needs. However, this is not exactly what we aimed to do. Dr. Sarver continued to highlight to us that the most important thing is meeting the students'

needs and showing them that we are designing **for** them. Right now, we have a good start, and there's definitely space to improve in the future, but we need to make that effort and ensure that the final product, after changes and adjustments, does have innovative features that accommodate specific student needs.

Biomedical Engineer (Dr. Dougherty):

Dr. Dougherty brought up similar points to Dr. Sarver. As previously mentioned, the final design wasn't all that our group had in mind. But Dr. Dougherty also reminded us about how important it is to stay creative. It is very easy to get caught up in getting the project done and focusing on having a product to present to the teachers. However, it is still really important to remember the creative side of engineering and consider that it takes a unique approach to create an effective solution. Overall, our group was able to take away that this time, we were way too focused on completing the project than making sure we were enjoying ourselves and creating out of the box solutions. As we make more iterations and adjust our design, our group will hopefully have less of a time strain and will be able to be thoughtful and have fun with our new solutions. In the future, we will also definitely keep this in mind because that's what engineering is all about.

Comprehensive Evaluation:

In order to address multiple aspects of our design/design process, we asked stakeholders and experts clear questions about specific parts we wanted reviewed. Safety and creativity were both important aspects of the design, so we made sure to ask questions like "What possible hazards of the design do you think could affect its safety?" and "How would you recommend we improve on the creativity of the design?"

External Evaluation:

Our external evaluations all conveyed a clear point. Our design was okay, but it could have been better. The whole point of this project was to create a design that catered to the specific needs of the special education classroom through LED lights. Our solution achieved half of this— we installed LED lights, but they weren't catered to any specific needs other than by providing zoning for the teachers. Overall, the project just fell short because we had a lot of ideas going into it and there was a lot of potential in general, but our approach was wrong. Our approach was so focused on getting the project done and being efficient, that we forgot a key part of the class: creativity and out-of-the-box thinking. We did a good job and the teachers are definitely going to get some use out of the lights. They were very excited when the lights were installed and felt satisfied with the outcomes. But as engineers, our job is to push ourselves to do better and create an even more effective product. So although our stakeholders are happy (that is the most important thing), we should be ambitious and look forward to doing even better next time. There is always room

for improvement and our group is excited to make the LED lights even better and continue to build upon our project in the future.